Mathematics GCE O Level 2022 - Paper 2

Duration: 2h30 | Cameroon GCE Board

This paper is arranged in two sections, A and B. Answer ALL questions in Sections A and B. Section A: Answer ALL the questions in the spaces provided (15 questions); the mark allocation for each question is indicated. Section B: All questions in Section B carry equal marks (15 marks each); the mark allocation for each question is indicated. You are reminded of the necessity for good English and orderly presentation in your answers. In calculations, you are advised to show all the steps in your working, giving your answer at each stage. Calculators are allowed. Duration: two and a half hours.

Exercise 1 (6 marks)

a. Express 6.093 to one decimal place.

b. Evaluate $\frac{2}{3} + \frac{1}{5} \div \frac{3}{4} - \frac{1}{3}$ *[6 marks for the whole question]*

Exercise 2 (6 marks)

Given the sets A, B and ξ defined as

$A = \{1, 2, 5, 7\}$

$B = \{2, 3, 6, 7\}$

$\xi = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$

Find

a. A^c

b. $A \cap B$

c. $n(B)$ *[6 marks for the whole question]*

Exercise 3 (5 marks)

Complete the truth table below

p	q	$\sim p$	$\sim p \vee q$
T	T	F	

p	q	$\sim p$	$\sim p \vee q$
T	F	F	
F	T	T	
F	F	T	

[5 marks]

Exercise 4 (6 marks)

The functions f and g are defined as

$f : x \mapsto x + 1$

$g : x \mapsto x^2$

Find

a. $g(-2)$

b. $f^{-1}(1)$

c. $gf(x)$ *[6 marks for the whole question]*

Exercise 5 (6 marks)

The floor of a rectangular room measures 600 cm long by 450 cm wide. It is to be covered with squared tiles of dimension 15 cm by 15 cm

a. Find the number of tiles used to cover the floor.

b. Given that each packet has 20 tiles,

(b) Calculate the number of packets needed *[6 marks for the whole question]*

Exercise 6 (5 marks)

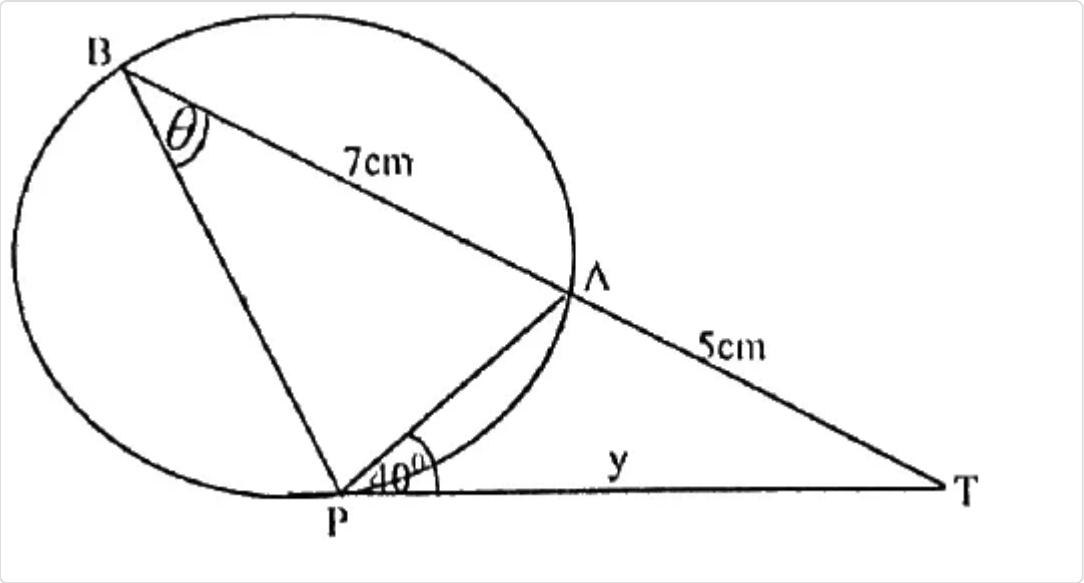


Figure 1 is a circle passing through ABP and a tangent PT. Given that angle $\mathrm{APT} = 40^0$, $\mathrm{AB} = 7\ \mathrm{cm}$ and $\mathrm{AT} = 5\ \mathrm{cm}$.

a. Find the angle marked θ

b. Calculate the value of y *[5 marks for the whole question]*

Exercise 7 (6 marks)

Given that the points A and B have coordinates $A(1, 2)$ and $B(3, 6)$.

Find :

a. the coordinates of the mid-point M of A and B

b. the gradient of the line segment AB

c. the gradient of the line perpendicular to AB *[6 marks for the whole question]*

Exercise 8 (6 marks)

Given that y varies directly as x^2, and $x = 1$ when $y = 3$

a. Form an equation in x and y

b. Find the value of y when $x = 4$ *[6 marks for the whole question]*

Exercise 9 (5 marks)

Given the equation $a = \dfrac{\sqrt{cr}}{3}$

a. Express r in terms of a and c

b. Find the value of r, when a=2 and c=3 *[5 marks for the whole question]*

Exercise 10 (6 marks)

Given that $(x-1)$ is a factor of $f(x) = x^2 + qx - 3$,

a. Find the value of q

b. With the value of q in (a), factorise $f(x)$

c. Hence, solve $f(x) = 0$ *[6 marks for the whole question]*

Exercise 11 (5 marks)

Given that the bearing of a ship, P, from a port, Q, is 080^0

a. Sketch a diagram to illustrate the movement

b. Find the bearing of Q from P *[5 marks for the whole question]*

Exercise 12 (6 marks)

Two matrices M and N are such that $M = \begin{pmatrix} 3 & 1 \\ 2 & 5 \end{pmatrix}$ and $N = \begin{pmatrix} a-1 & 1 \\ 2 & b+3 \end{pmatrix}$

Find

a. the values of a and b given that $M = N$

b. the transpose of M *[6 marks for the whole question]*

Exercise 13 (6 marks)

Given that the first term in an arithmetic progression (AP), is 3 and the fifth term is 19.

a. Determine the common difference

b. Find the first two terms of the AP

c. Find the sum of the first ten terms *[6 marks for the whole question]*

Exercise 14 (4 marks)

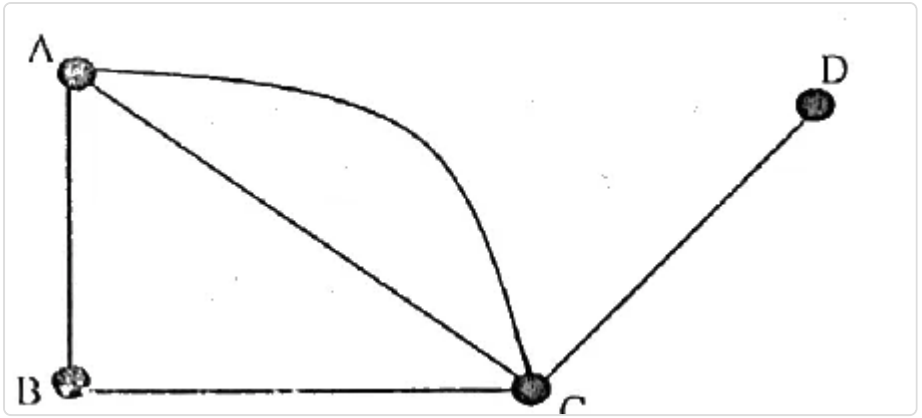


Figure 2 is a network A, B, C and D

State

a. the number of nodes

b. the number of edges

c. the number odd vertices *[4 marks for the whole question]*

Exercise 15 (5 marks)

A bag contains 3 ripe mangoes and 2 unripe mangoes. Two mangoes are drawn out of the bag at random one after another without replacement.

Find the probability that

a. the first mango is ripe

b. both mangoes are unripe *[5 marks for the whole question]*

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a. (i) A telephone dealer decides to import 200 smart phones at a total cost of 8 million FCFA. She pays 40% of the amount as taxes.

Find:

(a) the total amount spent on taxes.

b. her total expenditure on the phones

c. the cost of a phone

d. If she makes a profit of 15% on each phone,

Find

(d) the profit made on each phone

e. the total selling price of the phones *[10 marks for (i)]*

f. (ii) Solve the simultaneous equations

$2p - 3q = 12$

$3p + q = 7$ *[5 marks for (ii)]*

Using a scale of 2 cm for 1 unit on the x-axis and 1 cm for 1 unit on the y-axis, draw the graph of $y = x^2 - 3x - 4$ for $-2 \leq x \leq 5$.

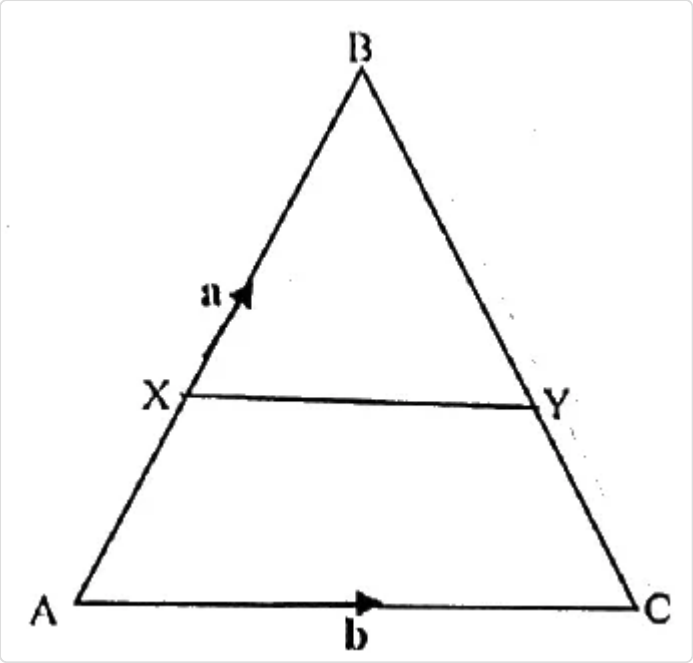
a. On the same graph paper, draw the line $y = 2x - 4$.

b. Using your graph,

(b) State the values of x for which $y = x^2 - 3x - 4$ cuts the x-axis.

c. State the minimum value of the curve $y = x^2 - 3x - 4$.

d. Solve the equation $x^2 - 5x = 0$. *[15 marks for the whole question]*



(The figure belongs to part (ii): in figure 3, X is the midpoint of AB and Y is the midpoint of BC, $\overrightarrow{AB} = \mathbf{a}$ and $\overrightarrow{AC} = \mathbf{b}$.)

a. (i) The marks obtained by 30 students in a Mathematics examnination are showed in the table below

3	**3**	**1**	**2**	**4**	**4**	**4**	**5**	**5**	**5**
3	3	3	5	5	7	7	7	8	8
7	7	7	10	10	9	9	10	9	10

(a) Draw up a frequency distribution table for the scores.

b. From the distribution table,

Find:

(b) The mode

c. The median

d. The mean

e. The probability that a student selected at random will score a mark less than 8. *[11 marks for (i)]*

f. (ii) In figure 3, X is the midpoint of AB and Y is the midpoint of BC. Given that $\overrightarrow{AB} = \mathbf{a}$ and $\overrightarrow{AC} = \mathbf{b}$

Find in terms of $\mathbf{a}$ and $\mathbf{b}$ the vectors

(a) $\overrightarrow{AX}$

g. (b) $\overrightarrow{CY}$ *[4 marks for (ii)]*

Using a scale of 1 cm to represent 1 unit on each axis, for $0 \leq x \leq 9$ and $0 \leq y \leq 10$

a. Draw and label triangle XYZ with coordinates $X(3, 2)$, $Y(6, 2)$ and $Z(6, 5)$.

b. The transformation T is a translation that maps triangle XYZ to $X'Y'Z'$, where $T = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$

(b) Find the coordinates of $X'Y'Z'$

c. Draw triangle $X'Y'Z'$

d. Reflect triangle XYZ on the y-axis to give $X''Y''Z''$

e. Write down the coordinates of $X''Y''Z''$ *[15 marks for the whole question]*

Want to check your answers?

The full correction of this paper, with the method explained step by step, is available on EduKora: 15,000 FCFA per child until the exam, 12,000 FCFA for each additional child in the same family.

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